

Determining the Optical Properties of Turbid Media Using the Adding-Doubling Method

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Abstract

A method is described for finding the optical properties (scattering, absorption, and scattering anisotropy) of a slab of turbid material using total reflection, unscattered transmission and total transmission measurements. This method is applicable to homogeneous turbid slabs with any optical thickness, albedo, or phase function. The slab may have a different index of refraction from its surroundings and may or may not be bounded by glass. The optical properties are obtained by iterating an adding-doubling solution of the radiative transport equation until the calculated values of the reflection and transmission match the measured ones. Exhaustive numerical tests show that the intrinsic error of the method is less than 3% when four quadrature points are used.

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1 Introduction

This paper introduces a practical way to determine the optical properties of scattering and absorbing materials. These properties are obtained by repeatedly solving the radiative transport equation until the solution matches the measured reflection and transmission values. The advantage over existing methods are increased accuracy and flexibility in modelling turbid samples with intermediate albedos, mismatched boundary conditions, and anisotropic scattering. The primary disadvantage is that this method is entirely numerical. For brevity, this method is called inverse adding-doubling (IAD): “inverse” implying a reversal of the usual process of calculating reflection and transmission from optical properties, and “adding-doubling” to indicate the method used to solve the radiative transport equation. The IAD algorithm and theory are described in this paper: an experimental implementation is presented in the companion paper [1].

The optical properties of biological tissue are important for photodynamic therapy and diagnostic techniques [2]. Typically, optical properties are obtained using solutions of the radiative transport equation that express the optical properties in terms of readily measurable quantities [3]. These solutions are either exact or approximate and correspond to the direct or indirect methods described by Wilson *et al.* [4]. Direct methods place stringent constraints on the sample to match the assumptions made for exact solution. For example, the direct method used by Flock *et al.* required very thin samples in which multiple scattering could be ignored [5]. Indirect methods relax the sample constraints but require approximations that are often invalid for tissue samples (e.g., nearly isotropic scattering or no internal reflection at the boundaries). The theory used in indirect methods usually falls into one of three categories: Beer’s law, Kubelka-Munk, or the diffusion approximation.

Beer’s law neglects scattering and is inappropriate for thick scattering materials. The Kubelka-Munk method and variants [6–12] are still used [13, 14], but are limited in their accuracy [11, 15]. Methods based on the diffusion approximation or a similar approximation (e.g., uniform radiances over the forward and backward hemispheres) tend to be more accurate than Kubelka-Munk [16, 17]. Techniques using the diffusion approximation include pulsed photothermal radiometry [18], time resolved spectroscopy [19], radial

reflectance spectroscopy [20], weak localization [21], and an iterative technique that uses reflection and transmission measurements [22]. These methods remain popular because they are easy to use, place relatively minor constraints on the type of sample, and are amenable to analytic manipulation. However, the diffusion approximation assumes that the internal radiance is nearly isotropic and consequently it is a poor approximation when scattering is comparable to absorption [23].

The adding-doubling method is a general, numerical solution of the radiative transport equation [24]. The adding-doubling method [25] was chosen because it is sufficiently fast that iterated solutions are possible on current microcomputers and sufficiently flexible that anisotropic scattering and internal reflection at the boundaries may be included. Other accurate solutions of the radiative transport equation such as Chandrasekhar's X and Y-functions [26], discrete ordinates [27], Monte Carlo [28–30], invariant embedding [31], or successive orders [24] are too slow or insufficiently flexible to incorporate the necessary boundary conditions needed for turbid materials with mismatched boundaries.

The IAD method consists of the following steps (1) Guess a set of optical properties; (2) Calculate the reflection and transmission using the adding-doubling method; (3) Compare the calculated values with the measured reflection and transmissions; and then (4) Repeat until a match is made. The set of optical properties that generates reflection and transmission values matching the measured values is taken as the optical properties of the sample. The results obtained using the IAD method are accurate for all optical properties and can be made arbitrarily precise at the cost of increased computation time. Furthermore, by avoiding an analytical solution, it is possible to incorporate the necessary corrections for measurements made with integrating spheres directly [32] (see Section 3.6). Such corrections are usually quite awkward to implement because the magnitude of the correction depends on the optical properties of the sample measured.

2 The Adding-Doubling Method

2.1 Definitions

The optical properties of a turbid medium are characterized by the absorption coefficient μ_a , the scattering coefficient μ_s , and the single-scattering phase function $p(\theta)$. The reciprocal of the absorption (scattering) coefficient is the average distance a photon will travel before being absorbed (scattered) by the medium. Two dimensionless quantities that characterize light propagation in a turbid medium are the albedo a and the optical thickness τ . These are defined as

$$a = \frac{\mu_s}{\mu_s + \mu_a} \quad \text{and} \quad \tau = d(\mu_s + \mu_a)$$

where d the physical thickness of the slab.

The single-scattering phase function $p(\theta)$ describes the amount of light scattered at an angle θ from the incoming direction. The phase function is often expressed in terms of the cosine of this scattering angle $\nu = \cos \theta$. The phase function is normalized so that its integral over all directions is unity

$$\int_{4\pi} p(\nu) d\omega = 2\pi \int_{-1}^1 p(\nu) d\nu = 1.$$

where $d\omega$ is a differential solid angle. The functional form of the phase function in tissue is usually not known. However work by Jacques *et al.* and Yoon *et al.* have shown that a Henyey-Greenstein function approximates single particle light scattering in human dermis and aorta at 633 nm [33, 34]. Consequently, this phase function will be used in all calculations in this paper

$$p(\nu) = \frac{1}{4\pi} \frac{1 - g^2}{[1 + g^2 - 2g\nu]^{3/2}}.$$

The Henyey-Greenstein phase function depends only on the anisotropy coefficient g , defined as

$$g = \int_{4\pi} p(\nu) \nu d\omega = 2\pi \int_{-1}^1 p(\nu) \nu d\nu.$$

Consequently, g is the average cosine of the scattering angle. When $g = 0$ then scattering is equally probable in all directions. Typical values for tissues in the red region of the spectrum are $g \approx 0.8$ [2], which is moderately forwardly directed.

Reflection and transmission are relative to the normal irradiance on the sample surface and vary between zero and one. The total reflection R_T is all light specularly reflected and backscattered by the sample. The total transmission T_T is all the light that passes through the sample and includes any light travelling through the sample without being scattered. This unscattered light is denoted T_C , because for collimated irradiance on a sample, the light passing directly through the sample is often called the collimated transmission. For a non-absorbing sample $R_T + T_T = 1$.

2.2 Theory

The doubling method was introduced by van de Hulst for solving the radiative transport equation in a slab geometry [35]. The advantages of the adding-doubling method are that only integrations over angle are required, physical interpretation of results can be made at each step, the method is equivalent for isotropic and anisotropic scattering, and results are obtained for all angles of incidence used in the integration [36]. The disadvantages are that it is (a) slow and awkward to calculate internal fluences, (b) suited to a layered geometry with uniform irradiation, and (c) necessary that each layer have homogeneous optical properties. For determining optical properties using only reflection and transmission, internal fluences are not needed so (a) is not a problem. Items (b) and (c) place restrictions on the sample geometry—the samples must be uniformly illuminated, homogeneous slabs. The adding-doubling method is well-suited to iterative problems because it provides accurate total reflection and transmission calculations with relatively few quadrature points. The method is very fast for small numbers of quadrature points, and consequently iteration is

practical.

In all calculations that follow the following assumptions are made: the distribution of light is independent of time, samples have homogeneous optical properties, the sample geometry is an infinite plane-parallel slab of finite thickness, the tissue has a uniform index of refraction, internal reflection at boundaries is governed by Fresnel's law, and the light is unpolarized. A non-absorbing layer with a different index of refraction may be present at the boundaries (glass slide).

The doubling method assumes that the reflection $R(\nu, \nu')$ and transmission $T(\nu, \nu')$ for light incident at an angle ν and exiting at an angle ν' is known for one layer. The reflection and transmission of a slab twice as thick is found by juxtaposing two identical slabs and adding the reflection and transmission contributions from each slab [24, 25]. The reflection and transmission for an arbitrary slab are calculated by first finding the reflection and transmission for a thin starting slab with the same optical properties (e.g., using single scattering) and then doubling until the desired thickness is reached. The adding method extends the doubling method to dissimilar slabs. Thus slabs with different optical properties can be placed adjacent to one another to simulate layered media or internal reflection caused by index of refraction differences.

Boundary conditions are implemented in the adding-doubling method by creating a layer which mimics the reflection and transmission at a boundary. This layer is added to a slab to find the reflection and transmission for a slab including the boundary conditions. If $r(\nu_i)$ is the unpolarized Fresnel reflection for light incident from a medium with index of refraction n_1 on a medium with index of refraction n_2 with a cosine of the angle as ν_i

$$R_{\text{bdry}}(\nu_i, \nu_j) = \frac{r(\nu_i)}{2\nu_i} \delta_{ij} \quad \text{and} \quad T_{\text{bdry}}(\nu_i, \nu_j) = \frac{1 - r(\nu_i)}{2\nu_i} \left(\frac{n_1}{n_0} \right)^2 \delta_{ij} \quad (1)$$

The square of the ratio of the indices of refraction is due to the n^2 -Law of radiance [37] which accounts for the difference in radiances across an index of refraction mismatch. The factor of $2\nu_i$ is included for uniformity with our original definition of the reflection function. If a glass slide is present then either two slabs should be used, or the equivalent reflection should be found by summing the infinite series. Both

operators are diagonal because light is specularly reflected and the angle of incidence equals the angle of reflection.

If a glass slide is present then the usual formulas that include multiple internal reflections are used. For example, if ν_i is the cosine of the angle of incidence from the turbid slab (index n_s) onto the glass slide (index n_g), then the cosine of the angle inside the glass slide (ν_g) is determined using Snell's law,

$$n_g \sqrt{1 - \nu_g^2} = n_s \sqrt{1 - \nu_i^2} \quad \nu_i < \nu_c$$

where ν_c is the cosine of the critical for total internal reflection. Defining $r_i = r(\nu_i)$ as the unpolarized Fresnel reflection coefficient for light passing from the slab into glass at an angle with cosine ν_i and $r_g = r(\nu_g)$ for light passing from glass to air, the net reflection coefficient that should be used above is

$$r(\nu) = \begin{cases} \frac{r_1(\nu_i) + r_g(\nu_g) - 2r_1(\nu_i)r_g(\nu_g)}{1 - r_1(\nu_i)r_g(\nu_g)}, & \text{if } \nu_i < \nu_c; \\ 1, & \text{if } \nu_i \geq \nu_c. \end{cases}$$

Since light is refracted at the boundary, care must be taken to ensure that the incident and reflected fluxes are identified with the proper angles.

The reflection and transmission functions for the thin starting layers were obtained using the diamond initialization method [38]. The optical thickness for starting layers was based on the smallest quadrature angle as suggested by Wiscombe [39]. The Henyey-Greenstein phase function was used for all calculations. Phase function renormalization was avoided by always using the δ - M method, which facilitates accurate calculations with highly anisotropic phase functions [40].

The adding-doubling method is based on numerical integration of functions using quadrature,

$$\int_0^1 f(\nu, \nu') d\nu' = \sum_{k=1}^N H_k f(x_k)$$

The values of the quadrature points x_k and the weights H_k are chosen in such a way that the integral is evaluated exactly for a polynomial of order $2N - 1$ (or possibly $2N - 2$ depending on the quadrature method).

Using N quadrature points (Gaussian) is equivalent to the spherical harmonic method of order P_{N-1} [41], i.e. four quadrature points corresponds to the P_3 method. The specific choice of quadrature methods for samples with mismatched boundaries is described in the next section.

2.3 Internal Reflection

Internal reflection at the boundaries (due to mismatched indices of refraction) was included in the calculation by adding an additional layer for each mismatched boundary. The reflection and transmission of this layer equaled the Fresnel reflection and transmission for unpolarized light incident on a plane boundary between two transparent media with the same indices of refraction. By sandwiching a tissue sample between glass or quartz plates, the usual irregularities in the tissue surface are minimized, and Fresnel reflection is a good approximation. The transmission and reflection of the boundary layer were zero and one respectively, for light incident at angles exceeding the critical angle for total internal reflection.

Total internal reflection caused problems by changing the effective range of integration. Usually, adding-doubling integrals range from 0 to 1, since the angle varies from $\frac{\pi}{2}$ to 0 and therefore the cosine varies from 0 to 1. The integrations are calculated using numerical quadrature, and the quadrature angles are optimized for this range. If the cosine of the critical angle is denoted by ν_c for a boundary layer with total internal reflection, then the effective range of integration is reduced to ν_c to 1 (because the rest of the integration range is now zero). To maintain integration accuracy, the integral is broken into two parts and each is evaluated by quadrature over the specified subrange,

$$\int_0^1 A(\nu, \nu') B(\nu', \nu'') d\nu' = \int_0^{\nu_c} A(\nu, \nu') B(\nu', \nu'') d\nu' + \int_{\nu_c}^1 A(\nu, \nu') B(\nu', \nu'') d\nu'.$$

Here $A(\nu, \nu')$ and $B(\nu, \nu')$ represent reflection or transmission functions, and clearly if either is identically zero for values of ν less than ν_c , the integration range is reduced. The calculations in this paper used Gaussian quadrature [42] for the range from 0 to ν_c , thereby avoiding calculations at both endpoints (in particular, the angle $\nu = 0$ is avoided, which may cause division by zero). Radau quadrature was used for

the range from ν_c to 1, because one quadrature angle may be specified [43]. Normal irradiance corresponds to $\nu = 1$, and if this angle is specified then interpolation between quadrature points is not needed to obtain reflection and transmission for normal unscattered irradiance. Interpolation can be a significant source of error. The number of quadrature points are divided evenly between 0 to ν_c and ν_c to 1, Since the quadrature methods work well with even numbers of quadrature points, this dictates that the quadrature points should be chosen in multiples of four.

3 The Iteration Process

The iteration process consists of finding optical properties that generate the measured reflection and transmission values. This section begins by showing that a unique solution to the inverse problem exists. The simplification necessary when one or more of the experimental measurements is missing are described next. Finally, three aspects of the iteration process are given (1) the function that defines the “distance” the calculated values are from the measure values, (2) the initial set of optical properties guessed, and (3) the algorithm used to minimize this function.

3.1 Uniqueness

The iteration method implicitly assumes that a unique combination of the albedo, the optical depth, and the anisotropy is determined by a set of reflection and transmission measurements. Clearly, if a sample is so thick that an accurate unscattered transmission measurement cannot be made then there are more unknowns than observations. Even when all the measurements are measureable, it is not obvious that a unique set of optical properties (a, τ, g) exists for any set of measurements (R_T, T_T, T_C) . For example, increasing the albedo of a sample will increase its total reflection and decrease its total transmission—but so will increasing the optical thickness of the sample. Uniqueness will be demonstrated for two cases: fixed unscattered transmission and fixed scattering anisotropy. The former is representative when (R_T, T_T, T_C) are all known; the latter applies when the unscattered transmission measurement is unavailable and a fixed

value for the scattering anisotropy must be assumed.

The dependence of the total transmission and total reflection on the anisotropy and albedo is shown in Figure 1. The bold (a, g) grid was computed with the unscattered transmission fixed at ten percent ($T_C = 0.1$) and the boundaries of the sample matched with its environment. The intersection of the measured total reflection and total transmission grid lines determines a *unique* albedo and anisotropy.

In Figure 2 the dependence of the total transmission and the total reflection on the reduced albedo and reduced optical thickness is shown. Scattering in the medium is assumed isotropic ($g = 0$) and the index of refraction is 1.4 [44]. Again, any non-zero reflection and transmission measurement yields a unique value for the reduced scattering and absorption coefficient. Figure 2 is very useful for obtaining quick estimates of the optical properties of a sample. For example, if $R_T = 0.4$ and $T_T = 0.2$, then $a \approx 0.96$ and $\tau \approx 7$. If the thickness of the sample is 0.4mm and the scattering anisotropy is assumed to be ≈ 0.8 , then using the similarity relations 3 and 4 the absorption and scattering coefficients will be $(\mu_a, \mu_s, g) \approx (0.7\text{mm}^{-1}, 17\text{mm}^{-1}, 0.8)$

3.2 Simplification

The three measurements usually available are the total reflection, the total transmission, and the unscattered transmission. It is assumed that the unscattered reflection from each surface is equal to the Fresnel reflection for light normally incident on the sample. From the unscattered transmission T_C , the optical thickness τ may be immediately calculated by solving

$$T_C = \frac{(1 - r_{s1})(1 - r_{s2})e^{-\tau}}{1 - r_{s1}r_{s2}e^{-2\tau}}, \quad (2)$$

where r_{s1} and r_{s2} are the primary reflections for light normally incident on the front and back surfaces of the slab. If glass slides are present the specular reflection coefficients r_{s1} and r_{s2} should include multiple internal reflections in the slide. Once the optical thickness is known only the albedo and anisotropy remain to be determined. These are varied until the calculated reflection and transmission match the measured values.

When the only measurements available are the total reflection and transmission, as might happen for a thick sample in which an accurate unscattered transmission measurement cannot be made, then only two optical parameters can be determined. Typically, a value for the scattering anisotropy is assumed, and the albedo and optical thickness are calculated based upon this assumed value. The accuracy of, say, fluence calculations using these values is unknown.

If only one measurement is available, then the sample is usually too thick to make a transmission measurement. In this case, the optical thickness of the sample is assumed infinite and, again a fixed value for the scattering anisotropy is chosen. The reduced albedo can now be calculated. This is a simple one parameter minimization problem and convergence is robust and rapid.

3.3 Metric

When the reflection and transmission for a particular set of optical properties are calculated a definition of how far these values are from the desired ones is needed. Several metrics were tried, but one based on a sum of relative errors worked best,

$$M(a, \tau, g) = \frac{|R_{\text{calc}} - R_{\text{meas}}|}{R_{\text{meas}} + 10^{-6}} + \frac{|T_{\text{calc}} - T_{\text{meas}}|}{T_{\text{meas}} + 10^{-6}}$$

here R_{meas} and T_{meas} are the measured reflection and transmission for scattered light. The factor of 10^{-6} is included to prevent division by zero when the reflection or transmission is zero. Note that the measurement errors for reflection and transmission are assumed equal for this metric. If this is not true then suitable modifications are necessary.

3.4 Inverse Method

The iteration method uses an N -dimensional minimization algorithm based on the downhill simplex method of Nelder and Mead [45]. One implementation of this method (**AMOEBA**) varies the parameters from $-\infty$ to ∞ [42]. Since the anisotropy, albedo, and optical depth have fixed ranges, they are transformed into

a “computation space.” For example, the transformation function for the albedo is

$$a_{\text{comp}} = \frac{2a - 1}{a(1 - a)},$$

Thus, as a_{comp} varies from $-\infty$ to ∞ , the albedo a varies from 0 to 1. The transformation for the anisotropy g (which varies from -1 to 1) is

$$g_{\text{comp}} = \frac{g}{1 + |g|},$$

The transformation for the optical thickness τ (that varies from 0 to ∞) is

$$b_{\text{comp}} = \ln(\tau)$$

These relations are easily inverted to obtain relations for the optical properties in terms of the computation values. All calculations were made using the real values: the transformed values were only used by the simplex method to chose the next iteration point. Typical convergence was in 20-30 iterations. Local minima exist, and convergence to such points is automatically detected because the magnitude of such minima exceeds the convergence criterion. A few different initial points may be necessary before the global minimum is found.

3.5 Initial Values

The initial set of optical properties chosen affects both the rapidity of convergence and the convergence to the correct values. Clearly, with a better first guess, fewer subsequent iterations will be needed. Poor guesses have the added disadvantage that the minimization algorithm may converge to a relative rather than the global minimum. Since the global minimum corresponds to the unique solution described above, relative minima must be guarded against. Fortunately, relative minima are easily detected by examining the magnitude of the minimum. If the magnitude exceeds a small tolerance then the minimum is incorrect and the iteration process must be restarted with a better initial guess.

Generating an initial set of optical properties that are roughly correct for any reflection and transmission

combination is not simple. Most approximations are accurate only for limited ranges of optical properties and fail badly when values outside this range are used.

Often it is convenient to use the similarity relations [46] to reduce the number of optical properties necessary to describe light propagation behavior. These approximate relations relate (a, τ, g) to the reduced optical properties $(a', \tau', g' = 0)$,

$$a' = \frac{a(1-g)}{1-ag} \quad \text{and} \quad \tau' = (1-ag)\tau. \quad (3)$$

The inverse relations are

$$a = \frac{a'}{1-g+a'g} \quad \text{and} \quad \tau = \tau' + \frac{a'\tau'g}{1-g}. \quad (4)$$

These sets $(a', \tau', 0)$ and (a, τ, g) will have similar total reflection and total transmission values.

For example, originally the Kubelka-Munk relations were used as values for initial guesses [47]. Unfortunately, these values were often worse than just using a fixed guess $(a', \tau', g) = (.5, 1, .2)$ to begin all iterations. A better starting set of starting values is based on a crude fit of the reflection and transmission values of Figure 2. The formula for the reduced albedo is

$$a' = \begin{cases} 1 - \left(\frac{1 - 4R_d - T_T}{1 - T_T} \right)^2 & \text{if } \frac{R_d}{1 - T_T} < 0.1 \\ 1 - \frac{4}{9} \left(\frac{1 - R_d - T_T}{1 - T_T} \right)^2 & \text{if } \frac{R_d}{1 - T_T} \geq 0.1. \end{cases}$$

The formula for the reduced optical thickness is

$$\tau' = \begin{cases} \frac{-\ln T_T \ln(0.05)}{\ln R_T} & \text{if } R_d \leq 0.1 \\ 2^{1+5(R_d+T_T)} & \text{if } R_d > 0.1. \end{cases}$$

Once τ has been obtained using Equation 3, and a' and τ' have been calculated as above, then Equations 1 and 2 can be used to generate a single set of starting values (a, τ, g) .

3.6 Integrating Sphere Correction

Total transmission and reflection are usually measured with integrating spheres. Secondary effects arising from interaction between the sample and the spheres make the integrating sphere signal no longer proportional to the sample reflection or transmission [32]. For example, when a double integrating sphere arrangement is used, the power on the wall of the reflectance sphere P_r normalized to the incident power P is [32]

$$\frac{P_r}{P} = \frac{\gamma_1(R_c(1 - \gamma_2 R_d) + \gamma_3 R_{cd})(1 - \gamma_4 R_d) + \gamma_1 \gamma_3 \gamma_4 T_d(T_{cd} + \gamma_5 T_C)}{(1 - \gamma_4 R_d)(1 - \gamma_6 R_d) - \gamma_4 \gamma_6 T_d^2},$$

where the γ_i 's depend only on the geometry and reflectivity of the integrating spheres, and various R 's and T 's correspond to different types of reflection and transmission by the sample. A similar formula holds for the normalized power on the transmission sphere wall. Once the integrating spheres have been characterized (i.e., the γ_i 's are known), the normalized powers on the walls of the integrating sphere can be calculated using the various R 's and T 's available using the adding-doubling method. If the inverse adding-doubling method is now modified to match the measured and calculated powers in the integrating spheres, then no conversion of the integrating sphere measurements to reflection and transmission must be made.

4 Accuracy of the Method

This section addresses the question: If R_T , T_T , and T_C are known *exactly*, what is the maximum possible error in the derived optical properties caused by the inverse adding-doubling method? The error analysis must be made numerically, because analytical expressions for light propagation in anisotropic media with mismatched boundaries are not available. The numerical tests are designed to find the accuracy of the inverse algorithm as functions of both the optical properties and the reflection and transmission of the sample. Implicit are checks on (a) the convergence of the inverse algorithm to the global minimum, (b) the termination criterion for stopping the iteration procedure, (c) the choice of starting parameters, and (d) the

effect of using small numbers of quadrature points in the algorithm? This last test is particularly important, since the method would be useless (too slow), if high accuracy could only be achieved using many quadrature points.

Generating a set of accurate testing values presented a problem, since tabulated accurate values for mismatched boundaries could not be found in the literature. Consequently, after demonstrating that our adding-doubling implementation reproduced the published values for matched boundaries [24, 46] exactly, we used Monte Carlo to verify a few mismatched cases. This confirmed the boundary condition algorithm and we then compared 36 and 48 quadrature point adding-doubling calculations. The 36 point calculation was always correct to at least 0.1%. The test data was constrained to include only samples with reduced optical thicknesses greater than $1/4$ and optical thicknesses less than 32. The rationale is that when $\tau' < 1/4$, either the sample does not multiply scatter and therefore a simpler algorithm should be used or the sample does multiply scatter but due to the highly anisotropic nature of each scattering event, separation of T_C from scattered light will be very difficult. The second criterion ($\tau < 32$) ensures the sufficient collimated transmitted light passes through the sample that it can be possibly measured ($T_C \sim e^{-32} \sim 10^{-14}$). The IAD method works for *any* optical thicknesses but it becomes progressively less accurate outside this range.

All samples assume slabs with an index of refraction of 1.4 relative to the environment. Results for the index matched samples indicate that the error in the IAD method is roughly half those presented here for the mismatched case. This is caused by total internal reflection of light at quadrature angles greater than the critical angle (one-half of the total number, see Section 2.3). The slabs are not sandwiched by glass: sandwiching the sample between glass slides did not alter the mismatched ($n = 1.4$) results significantly and are omitted for brevity.

In all calculations, the scattering phase function used was the Henyey-Greenstein phase function. Despite the limited experimental evidence that this phase function is appropriate for tissues, errors arising from an inappropriate phase function should be small, since different phase functions with equivalent g values and different higher order moments generate nearly equal reflection and transmission values [46].

The absolute error is defined as the difference between the “true” or accurate values and those calculated

using the inverse adding-doubling method,

$$\Delta a' = |a'_{\text{true}} - a'_{\text{calc}}|. \quad (5)$$

The relative percentage error is the absolute error divided by the true value and multiplied by 100. The absolute errors in the scattering and absorption coefficients are proportional to the physical thickness of the sample and consequently are not particularly useful. The relative errors for μ_a and μ_s can also be obtained, despite the fact that only dimensionless optical parameters are used by noting that the physical thickness cancels,

$$\frac{\Delta \mu_a}{\mu_a^{\text{true}}} = 100 \cdot \left[1 - \frac{(1 - a_{\text{calc}})\tau_{\text{calc}}}{(1 - a_{\text{true}})\tau_{\text{true}}} \right] \quad \text{and} \quad \frac{\Delta \mu_s}{\mu_s^{\text{true}}} = 100 \cdot \left[1 - \frac{a_{\text{calc}}\tau_{\text{calc}}}{a_{\text{true}}\tau_{\text{true}}} \right]. \quad (6)$$

4.1 Variation with optical properties

The sensitivity of the inverse adding-doubling method to the optical properties of a sample was evaluated using a data set consisting of reflection and transmission values for eleven reduced albedos spaced such that $\sqrt{1 - a'_{\text{true}}} = (0, 0.1, \dots, 1)$, ten anisotropies spaced similarly $(0, 0.19, 0.36, 0.51, 0.64, 0.75, 0.84, 0.91, 0.96, 0.99)$, and the fourteen reduced optical depths (listed in Table 1). Thus for each pair (a', τ') there were ten different values of g that all corresponded to roughly equal values of (R_T, T_T) . A total of 1540 different combinations of $(a_{\text{true}}, \tau_{\text{true}}, g_{\text{true}})$. For each combination, accurate reflection and transmission (R_T, T_T, T_C) values were calculated. The optical properties were found using the IAD method with four and eight quadrature points. Diffusion equation results were obtained using the same iteration algorithm, but by replacing the adding-doubling calculation with a delta-Eddington diffusion approximation [22, 48].

The spacing in the reduced albedo was based on the observation that reflection and transmission are very sensitive to changes in the reduced albedo when it is near unity [24]. The anisotropy was chosen to vary in a similar manner, for the same reason. Completely forward scattering ($g = 1$) was omitted because it corresponds to the non-scattering case. The non-scattering ($a = 0$) and non-absorbing ($a = 1$) cases were both included. The range of the reduced optical thickness was chosen because of the natural advantage

obtained by making calculations for optical thicknesses that vary by a factor of two using the adding-*doubling* technique.

The variation in IAD accuracy with reduced optical thickness is tabulated in Table 1 and graphed in Figure 3. The maximum relative error for a fixed τ' is the greatest error in a calculated value of τ' for *any* albedo and scattering anisotropy. The relative error decreases as the number of quadrature points increases. When $1 \leq \tau' \leq 16$, the maximum relative error using four quadrature points is less than 2%. Surprisingly, the diffusion approximation works best for $\tau' \approx 1$, but only when just the relative error is considered. Table 1 shows that the maximum *absolute* error $\Delta\tau'$ monotonically increases with τ' for the diffusion approximation as well as four and eight quadrature point IAD calculations.

The variation in the relative error in the scattering coefficient was evaluated for changes in the reduced optical thickness, the reduced albedo, and in the scattering anisotropy using only four quadrature points. In Figure 4, the maximum relative error in the calculated scattering coefficient is plotted as a function of the three dimensionless quantities a' , τ' , and g . As expected, the accuracy in determining μ_s increases as the scattering increases (i.e., as $a' \rightarrow 1$). Increased scattering implies more uniform internal radiance distributions, which in turn, are more accurately approximated by four quadrature points than highly anisotropic internal radiance distributions. Errors in the scattering coefficient do not depend strongly on either the reduced optical thickness or the scattering anisotropy. The maximum relative error never exceeds 6% for positive albedos.

4.2 Variation with Reflection and Transmission

The accuracy of the inverse adding-doubling method varies with the reflection and transmission of the sample. To quantify the maximum error for specific reflection and transmission values another paired set of $(a_{\text{true}}, \tau_{\text{true}}, g_{\text{true}})$ and (R_T, T_T, T_C) were generated. In this set the total reflection and total transmission were evenly spaced (in 0.05 increments). By allowing the unscattered transmission to take on different values, ten different sets of $(a_{\text{true}}, \tau_{\text{true}}, g_{\text{true}})$ were obtained for each pair (R_T, T_T) . The maximum relative error in the absorption coefficient, the scattering coefficient, and the scattering anisotropy was calculated using the

IAD method with four quadrature points.

Figure 5 shows a contour plot of the maximum relative error in the calculated absorption coefficient as a function of total reflection and total transmission. The maximum relative error in the absorption coefficient (assuming $\mu_a > 0$ or $R_T + T_T < 1$) was 10%. The case $R_T + T_T = 1$ is omitted because it $\mu_a = 0$ for conservative scattering and the relative error is infinite in this case. For the majority of (R_T, T_T) combinations, the maximum error is 2-3%. The error is greatest when the total transmission highest, which corresponds to samples that absorb only a small fraction of the light.

Figure 6 shows a contour plot of the maximum relative error in the scattering coefficient as a function of total reflection and total transmission. In contrast to the error in the absorption coefficient, the error for the scattering coefficient is greatest when little light is reflected by the sample. The error drops sharply with increasing reflection value. The maximum error is always less than 2% for any non-zero value of μ_s . The case $\mu_s = 0$ is omitted for the same reason conservative scattering is avoided in Figure 5. The larger relative errors seen in Figure 4 all correspond to very small reflection values indicated by the black area.

Figure 7 shows a contour plot of the maximum absolute error in the scattering anisotropy as a function of total reflection and total transmission. The error is always less than 0.03 for all cases in which the scattering coefficient is non-zero. When the scattering coefficient is zero, then no scattering takes place and any measurement will not give information on the shape of the scattering event. These cases correspond to total reflection values which are equal to the unscattered reflected light.

Typically, the errors due to the IAD method will be much smaller those due to experimental uncertainties. For example, assume the measurements $R_T = 0.264$, $T_T = 0.261$, and $T_C = 5.91 \times 10^{-5}$ are known with a 1% uncertainty. The relative errors due to the IAD method are 0.04% for μ_s , 1% for μ_a , and 0.1% for g . However, perturbing the reflection and transmission values by 1% and re-solving for the optical properties results in errors of 0.4% for μ_s , 17% for μ_a , and 0.4% for g , or roughly an order of magnitude larger than the errors inherent in the IAD method itself. *Such error estimates are necessary* because of the non-linear relationship between reflection and transmission and the intrinsic optical properties. A primary goal of this work has been to develop an accurate method for computing optical properties, so that such estimates can

be made without introducing errors caused by the calculation itself.

In practice, the two most common sources of experimental errors are (1) loss of light out the edges of the sample (thereby invalidating the one-dimensional assumptions and underestimating both R_T and T_T) and (2) collecting scattered light in the unscattered transmission measurement (and thereby overestimating T_C). The first problem manifests itself in spuriously high absorption values and the latter case leads to optical properties that depend on the thickness of the sample. These problems are discussed at length in the experimental implementation of this technique [1].

5 Conclusions

The inverse adding-doubling method translates total reflection, total transmission, and unscattered transmission measurements into scattering, absorption and scattering anisotropy values. The method is applicable when the one-dimensional radiative transport equation adequately describes the propagation of light through the sample. Using only four quadrature points, the IAD method generates optical properties (μ_a, μ_s, g) that are accurate to 2-3% for most reflection and transmission values. Higher accuracy is achieved using more quadrature points, but requires increased computation time. The validity of the IAD method for samples in which $(\mu_a \approx \mu_s)$ is especially important since other methods (e.g., those based on the diffusion approximation) are known to fail in this regime. Furthermore, both anisotropic phase functions and Fresnel reflection at boundaries are accurately approximated, and therefore the IAD method is well-suited to measurements involving biological tissues sandwiched between glass slides. An computer implementation of the inverse adding-doubling method is available from the first author.

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τ'	Maximum Absolute Errors			Maximum Relative Errors		
	Diffusion	4 Quad. pts	8 Quad. pts	Diffusion	4 Quad. pts	8 Quad. pts
1/4	.04	.02	.01	16	9	4
3/8	.03	.02	.01	9	6	3
1/2	.03	.02	.01	5	5	2
3/4	.03	.03	.01	3	3	1
1	.04	.03	.01	4	3	1
3/2	.07	.03	.01	5	2	1
2	.2	.04	.02	11	2	.8
3	.2	.04	.02	7	1	.6
4	.3	.05	.02	8	1	.6
6	.6	.1	.04	10	1	.6
8	.9	.2	.05	11	2	.6
12	1.5	.3	.09	12	2	.7
16	2	.4	.1	13	2	.7

Table 1: Maximum errors in the calculated reduced optical thickness τ' for any albedo or anisotropy as a function of the true reduced optical thickness. The calculated values were obtained using a delta-Eddington approximation (diffusion) or the inverse adding-doubling method with four or eight quadrature points

Figure Captions

Figure 1: Total reflection and total transmission of an index matched slab ($n = 1$) as a function of the albedo a and anisotropy g for a fixed unscattered transmission value of 10%. Each point on the bold (a, g) grid corresponds to a unique (R_T, T_T) pair.

Figure 2: Total reflection and total transmission of a slab as a function of the reduced albedo a' and reduced optical thickness τ' . Isotropic scattering is assumed as well as an index of refraction mismatch ($n = 1.4$). Each point on the (a', τ') grid corresponds to a unique (R_T, T_T) pair.

Figure 3: Maximum error in the calculated reduced optical thickness τ' for any anisotropy and any albedo using the diffusion approximation and the IAD method with four and eight quadrature points.

Figure 4: Maximum relative error in the calculated scattering coefficient as a function of the reduced albedo a' , reduced optical thickness τ' , and the scattering anisotropy g using the IAD method with four quadrature points and mismatched boundaries $n = 1.4$.

Figure 5: The relative error in the calculated absorption coefficient as a function of reflection and transmission. The IAD method was used with four quadrature points and the slab had mismatched boundaries.

Figure 6: The relative error in the calculated scattering coefficient as a function of reflection and transmission. The IAD method was used with four quadrature points and the slab had mismatched boundaries.

Figure 7: The relative error in the calculated scattering anisotropy as a function of reflection and transmission. The IAD method was used with four quadrature points and the slab had mismatched boundaries.

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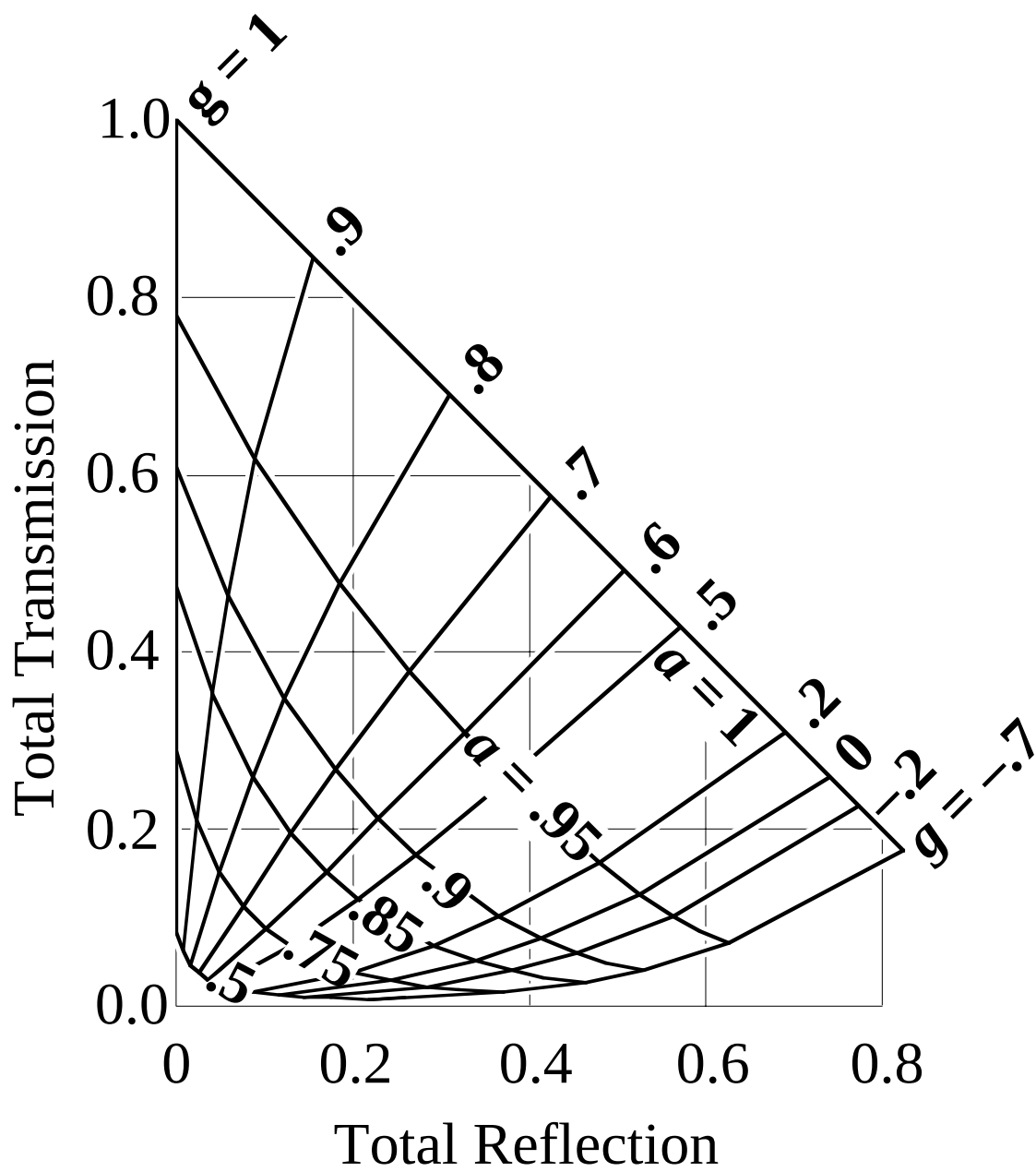


Figure 1

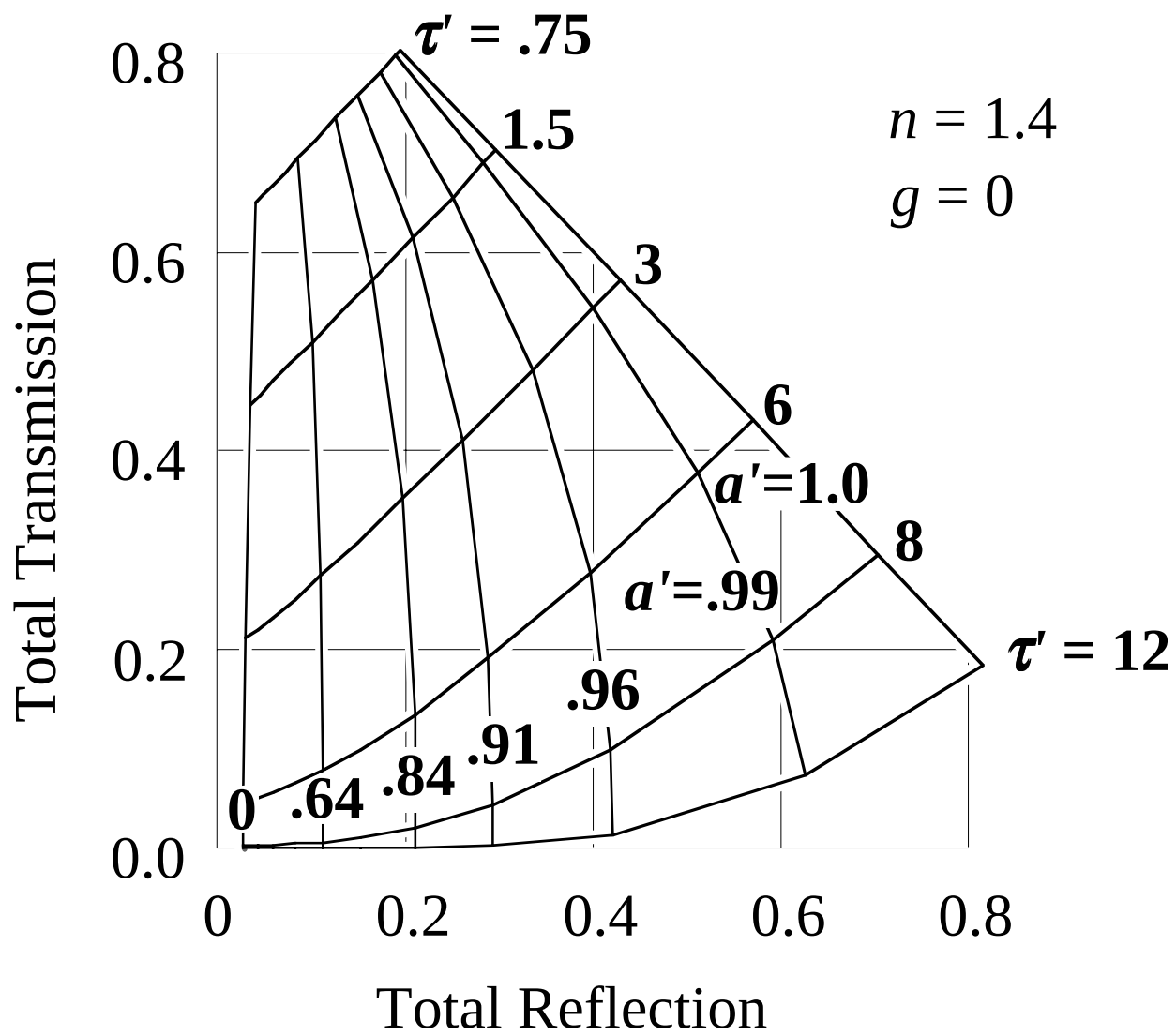


Figure 2

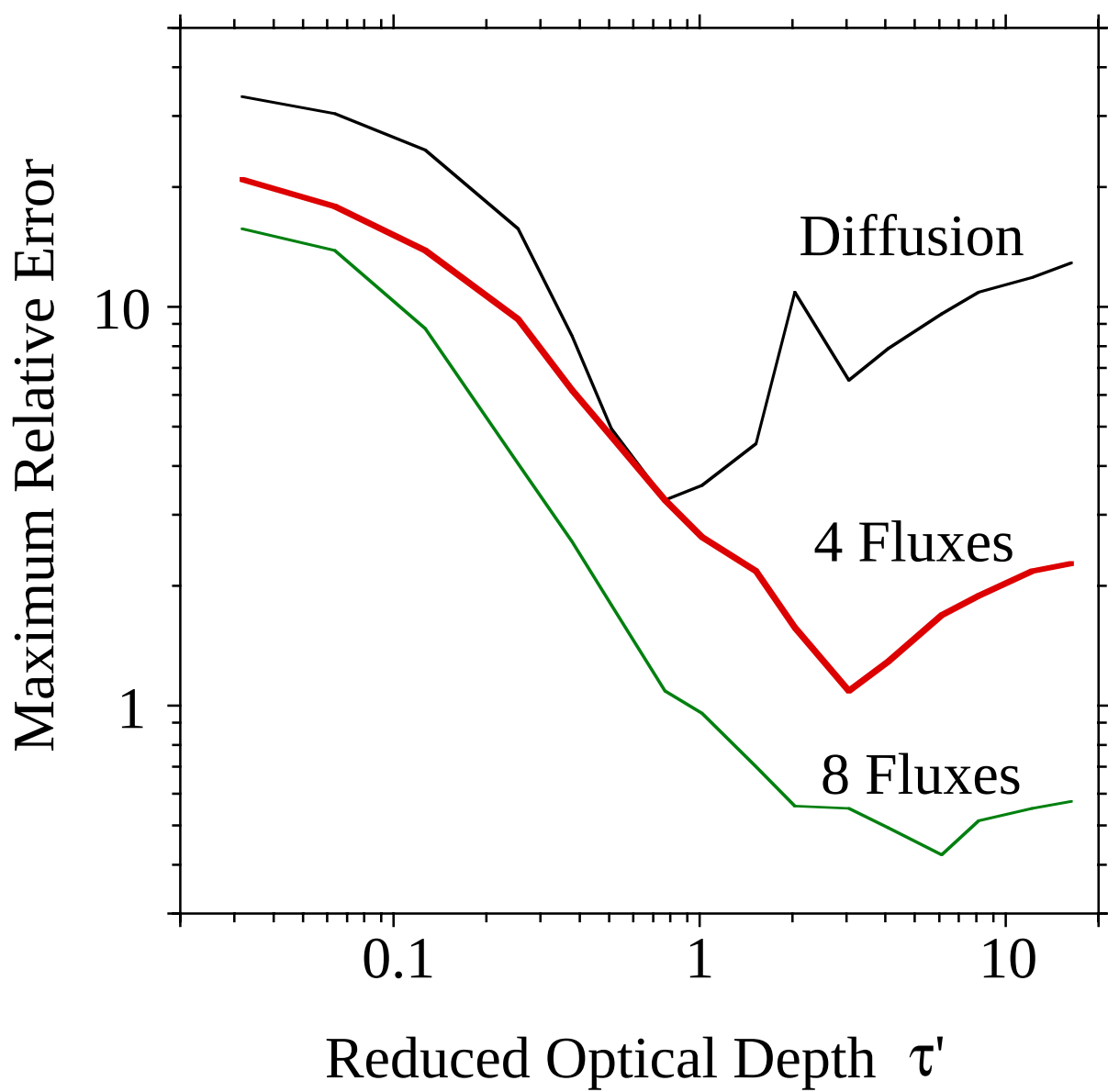


Figure 3

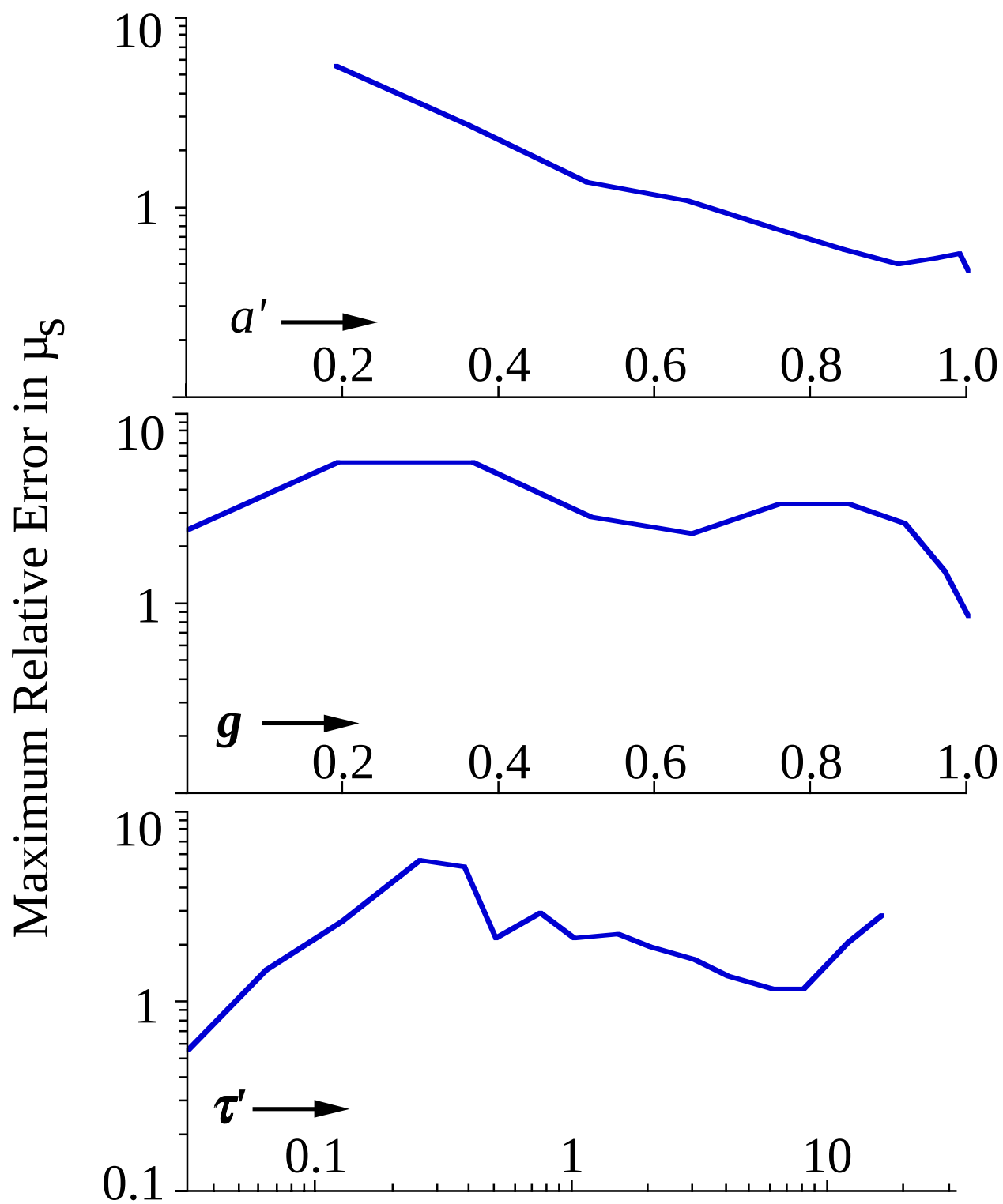


Figure 4

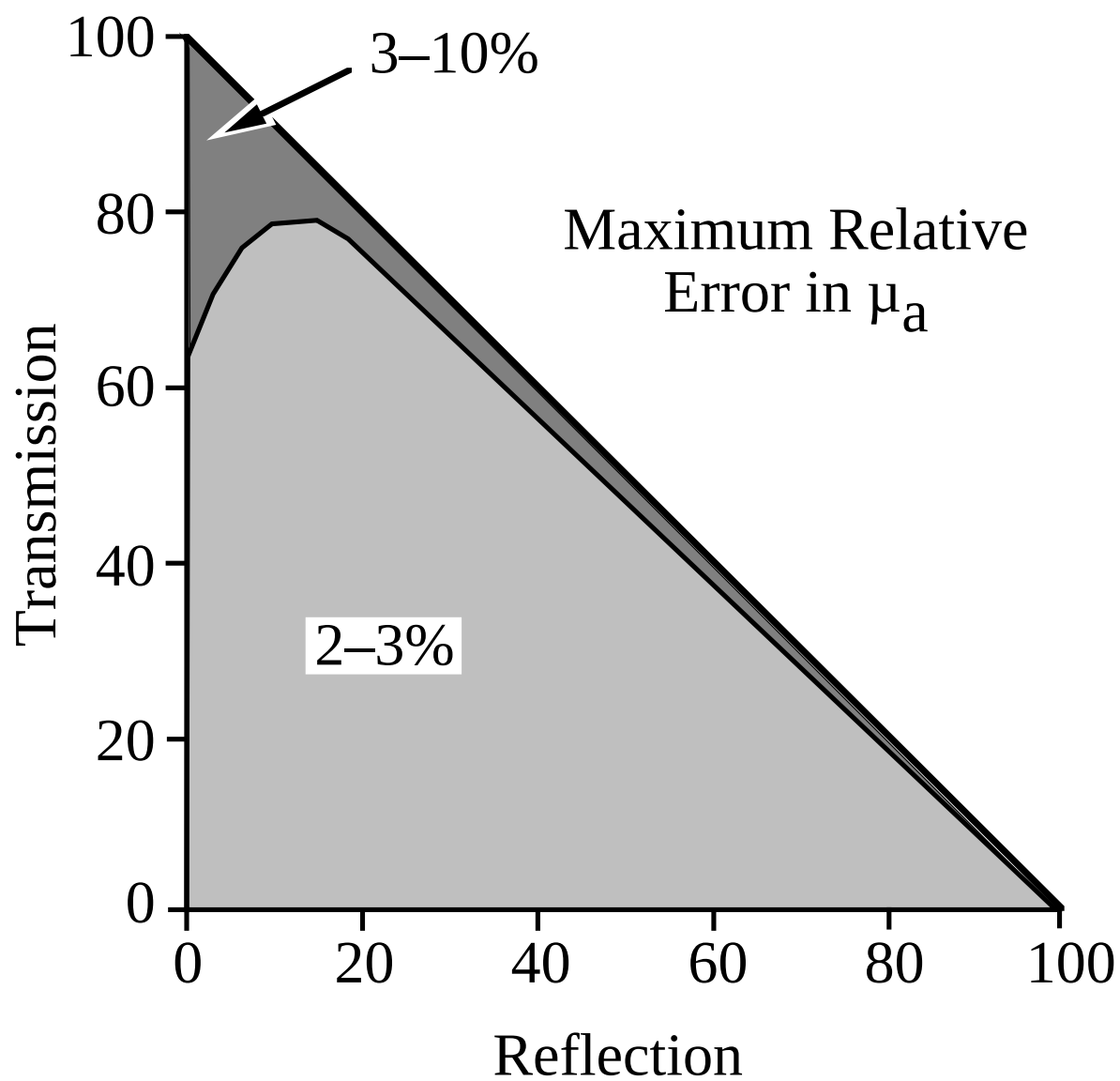


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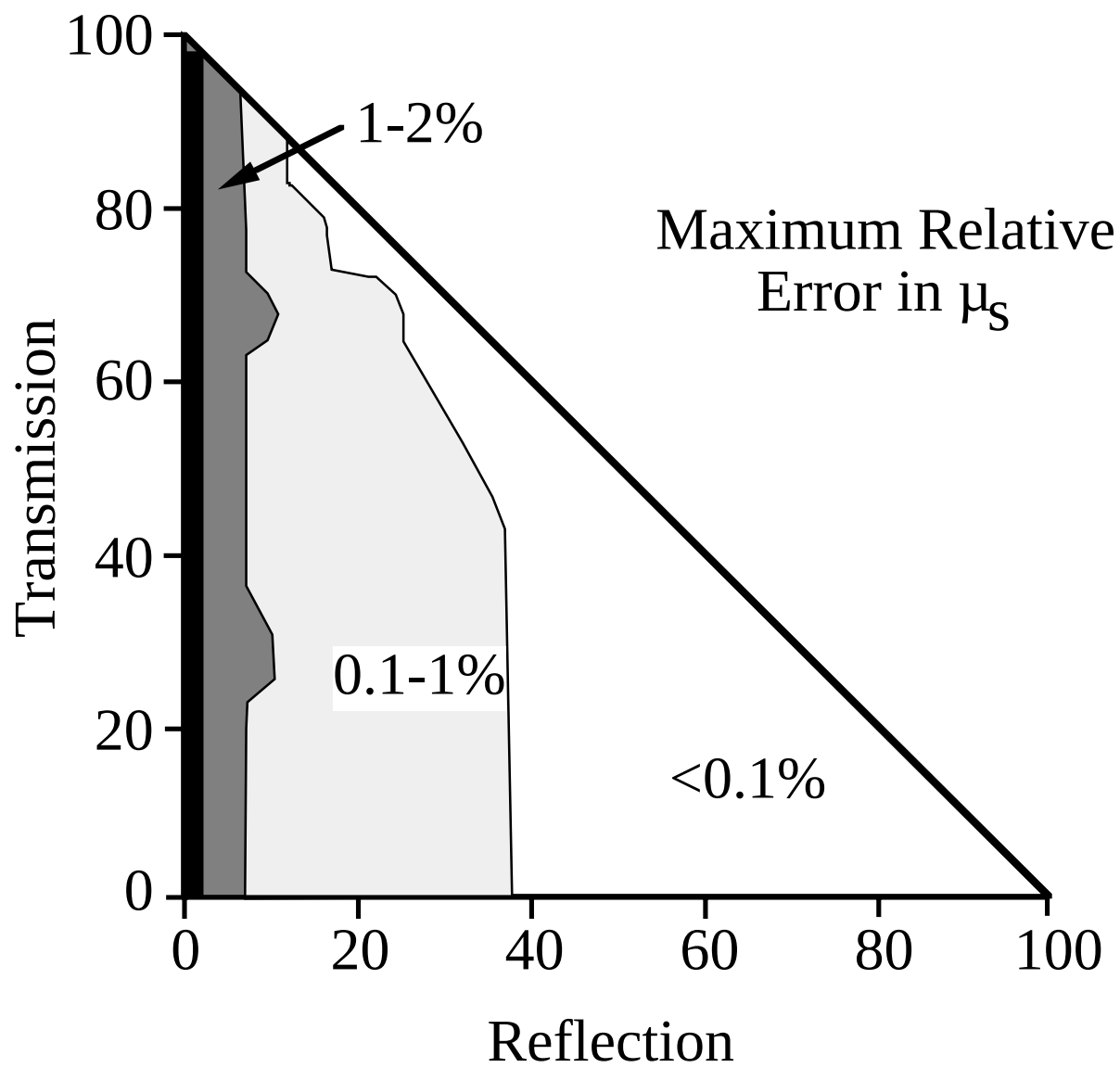


Figure 6

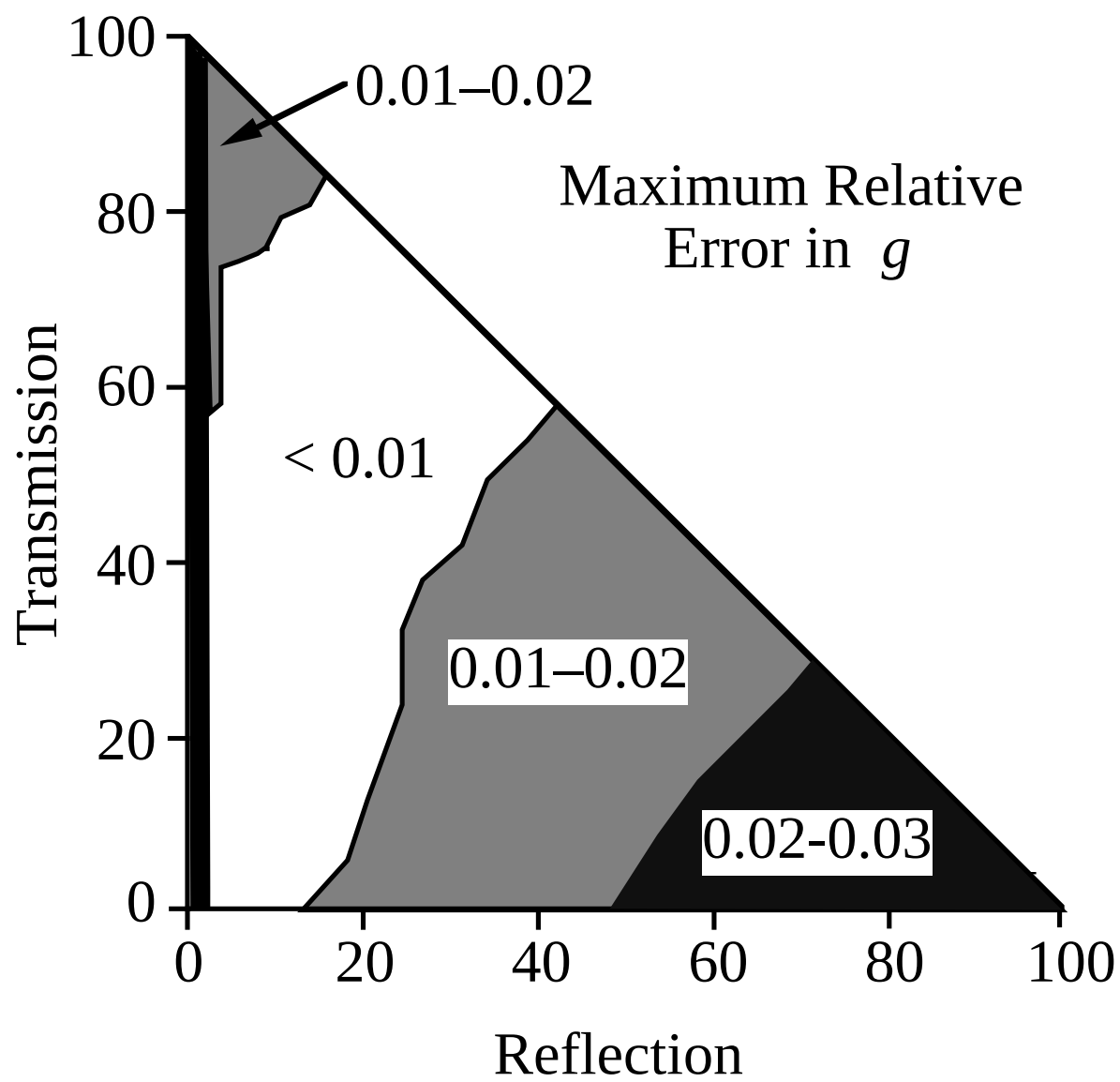


Figure 7